U.S.T.H.B. / Faculty of Mathematics — Academic year: 2026/2027
Department of Operations Research — Course: Thor (S1, bigDataA)

Exercise Sheet No. 2 (Scheduling on a Single Machine)

Exercise 1: Given the table below, solve the problem $1//\overline{C}$.

T_i	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8
p_i	10	2	4	7	3	1	3	2

Exercise 2: Given the table below, solve the problem $1//L_{max}$.

T_i	T_1	T_2	T_3	T_4	T_5	T_6	T_7
p_i	29	13	31	20	7	3	9
d_i	80	20	67	48	100	30	50

Exercise 3: Consider the following instance with three tasks and one machine:

T_i	T_1	T_2	T_3
p_i	2	1	1
d_i	3	6	5

1/ Compute L_{max} for the sequence T_1, T_2, T_3.

2/ Conclude.

Exercise 4: Consider the following instance with one machine and 7 tasks:

T_i	T_1	T_2	T_3	T_4	T_5	T_6	T_7
p_i	6	18	12	10	10	17	16
d_i	8	42	44	24	90	85	68

1/ Find an optimal schedule for the criterion T_{max}.

2/ Find an optimal schedule for the criterion $\overline{U}$.

Exercise 5: Given the table below, solve the problem $1//N_T$.

T_i	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9	T_{10}
p_i	5	3	1	2	4	4	2	1	1	4
d_i	19	16	25	3	8	14	31	23	2	15

Exercise 6: Different paint colours are produced in sequence on the same equipment, which must be thoroughly cleaned whenever the colour changes. The cleaning times (in hours) are as follows:

		Next colour				
		White	Yellow	Red	Violet	Blue
Previous colour	White	-	1	2	3	4
	Yellow	6	-	1	2	3
	Red	8	6	-	1	2
	Violet	10	8	6	-	1
	Blue	12	10	8	6	-

The production times (in hours) of the different paints are:

	Colour				
	White	Yellow	Red	Violet	Blue
Production time	13	14	15	16	15

The problem is to find the total production time. (Explain the complexity of the problem, the solution method and the solution found.)

Exercise 7: Seven tasks must be processed on a given machine. The following information is known about these tasks: (a) they are all currently available; (b) their processing times and the dates by which they must be delivered are given in the following table:

Task	1	2	3	4	5	6	7
Duration in days	4	5	3	6	7	2	1
Delivery due date	8	10	15	10	17	20	14

1/ Find a schedule that minimises the mean flow time (average time spent in the system) of these tasks. Give the order of the tasks and the value of the objective at the optimum.

2/ Find a schedule that minimises the largest lateness. Give the order of the tasks and the value of the objective at the optimum.

Exercise 8: A steel plant makes 5 products on a rolling mill. The processing time of each product and its delivery date are given in the table below.

Product number	1	2	3	4	5
Rolling time (min)	50	150	80	200	30
Delivery date (min)	100	300	410	400	200

1/ Suppose the products are processed in the order 3, 4, 1, 5, 2. Give a schedule and the mean completion time.

2/ Aiming to optimise the mean completion time, you must propose to this company the order in which to process the jobs. Give the optimal schedule and the optimal mean completion time.

3/ Suppose the products are processed in the order 5, 1, 3, 2, 4. Give a schedule, the minimum lateness, the maximum lateness and the mean lateness.

4/ Aiming to optimise the maximum lateness, you must propose to this company the order in which to process the products. Give the optimal schedule, the minimum lateness, the maximum lateness and the mean lateness.

5/ What does the mean lateness become if the order 1, 5, 2, 3, 4 is used? Conclude.